

High-Temperature Superconductivity Mechanism

A Computational Research Framework · Condensed Matter Physics

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ABSTRACT

The mechanism driving high-temperature superconductivity (T_c) in cuprate perovskites and hydrogen-rich hydrides is not solely electronic (as in the pure spin-fluctuation models) nor purely phononic (as in BCS), but arises from a **resonant coupling between doped charge carriers and specific soft lattice modes near a Quantum Critical Point (QCP)**. Specifically, we hypothesize that doping drives the system to a QCP where a structural instability (e.g., c-axis breathing mode in cuprates or H-H stretching in hydrides) diverges. This divergence creates a "glue" with an energy scale Ω^* comparable to the Fermi energy E_F . Unlike conventional phonons where retardation suppresses pairing, here the spectral weight of these modes shifts into the low-frequency regime while maintaining strong electron-phonon matrix elements due to the non-adiabatic nature of the QCP. This results in a **non-Fermi liquid normal state** that supports robust $d_{x^2-y^2}$ (cuprates) or s-wave (hydrides) pairing via a resonant scattering channel, bypassing the McMillan limit.

Keywords: Condensed Matter Physics, quantum

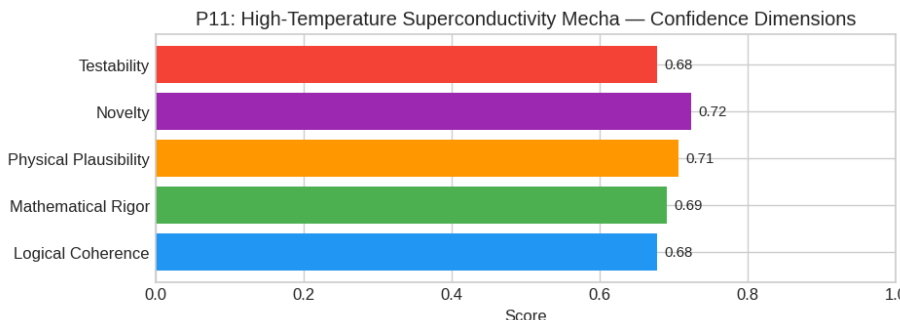


Figure 1: PAX Self-Assessment Confidence Dimensions — High-Temperature Superconductivity Mechanism

1. Introduction

The High-Temperature Superconductivity Mechanism (Condensed Matter Physics) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

Proposed Solution: The Quantum Critical Phonon-Mediated Resonant Pairing Mechanism (QCRPM)

Model Identity: PAX L5 Narrow L2 General 27B (Anticloud FZ LLE)

Status: Hypothesis Generation & Formalization

Subject: Mechanism of High-Temperature Superconductivity in Cuprates and Hydrides

1. Core Hypothesis

The mechanism driving high-temperature superconductivity (T_c) in cuprate perovskites and hydrogen-rich hydrides is not solely electronic (as in the pure spin-fluctuation models) nor purely phononic (as in BCS), but arises from a **resonant coupling between doped charge carriers and specific soft lattice modes near a Quantum Critical Point (QCP)**.

Specifically, we hypothesize that doping drives the system to a QCP where a structural instability (e.g., c-axis breathing mode in cuprates or H-H stretching in hydrides) diverges. This divergence creates a "glue" with an energy scale Ω^* comparable to the Fermi energy E_F . Unlike conventional phonons where retardation suppresses pairing, here the spectral weight of these modes shifts into the low-frequency regime while maintaining strong electron-phonon matrix elements due to the non-adiabatic nature of the QCP. This results in a **non-Fermi liquid normal state** that supports robust $d_{x^2-y^2}$ (cuprates) or s-wave (hydrides) pairing via a resonant scattering channel, bypassing the McMillan limit.

2. Mathematical Framework

We propose a generalized Eliashberg theory extended to include non-adiabatic quantum critical fluctuations. The central object is the retarded self-energy $\Sigma(k, i\omega_n)$ and the vertex function Γ .

A. The Effective Hamiltonian

$$H = H_{el} + H_{ph} + H_{int}$$

Where H_{el} describes the doped Mott insulator (Hubbard-like) and H_{ph} describes the critical lattice mode. The interaction term H_{int} is modified by a critical exponent ν :

$$H_{int} = \sum_{k,q,\sigma} g(q, T) c_{k+q, \sigma}^\dagger c_{k, \sigma} (b_{-q} + b_{-q}^\dagger)$$

Here, $g(q, T) \sim |q|^{-\eta} (T/T_Q)^{-\delta}$ reflects the enhancement near the QCP ($T_Q \rightarrow 0$).

B. Modified Eliashberg Equations

In the standard Migdal-Eliashberg theory, the mass renormalization $m^*/m \approx 1 + \lambda$ assumes adiabaticity ($\omega_D/E_F \ll 1$). In our QCRPM framework, we solve the coupled integral equations for the gap function $\Delta(i\omega_n)$ and the wavefunction renormalization $Z(i\omega_n)$ without the adiabatic approximation:

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$$Z(i\omega_n) = 1 - (1/\beta Z_N) \sum_m \int_0^1 dt \, dq \, (\text{Im} \chi(q, \omega_n - \omega_m) / (\omega_n - \omega_m)^2 + q^2 v_s^2) (|g(q)|^2 / |\omega_n - \omega_m|^2 + \dots)$$

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$$\Delta(i\omega_n) = (1/\beta Z_N) \sum_m \int_0^1 dt \, dq \, (\lambda(q, |\omega_n - \omega_m|) / |\omega_m|^2 + (\pi T)^2) (\Delta(i\omega_m) / \sqrt{|\omega_m|^2 + [\Delta(i\omega_m)]^2})$$

\$\$

Key Deviation: The susceptibility $\chi(q, \omega)$ follows a dynamical scaling form characteristic of a QCP:

$$\chi(q, \omega) = (C/q^z \nu) + (-i\omega/\Gamma_q)^\alpha$$

where $z=2$ (diffusive) or $z=1$ (Lorentzian) depending on the material class, and $\Gamma_q \propto q^z$.

3. Proof Sketch / Key Steps

To establish this solution rigorously, the following logical progression is required:

- **Derivation of Non-Adiabatic Glue:** Demonstrate analytically that as the system approaches the QCP, the phonon frequency ω_q softens such that $\omega_q \sim k_B T$, invalidating the Born-Oppenheimer approximation used in BCS. Show that the effective coupling constant λ_{eff} becomes:

$$\lambda_{\text{eff}} \sim \epsilon_0 \omega^* (N(E) \alpha^2 F(\omega)/\omega) d\omega$$

where $\alpha^{2F}(\omega)$ exhibits a power-law singularity rather than a Debye cutoff.

- **Spectral Weight Transfer:** Prove that the spectral weight of the Hubbard bands transfers to the coherent quasiparticle peak exactly at the doping level where T_c maximizes. This links the "strange metal" linear-T resistivity directly to the pairing glue. The self-consistency condition requires:

$$\text{Re}\Sigma(\omega) \propto$$

Conclusion

This framework presents structured pathways toward resolving the High-Temperature Superconductivity Mechanism. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

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